

## SCHORTENS JEVER, Germany

WMO: 101220

Lat: 53.534N

Lon: 7.889E

Elev: 7

StdP: 101.24

Time Zone: 1.00 (EUC)

Period: 94-11

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 12            | -8.8       | -6.9 | -11.0                         | 1.5 | -7.6 | -9.1 | 1.7 | -6.0 | 11.4                  | 7.2  | 10.5 | 5.8  | 3.4                   | 100  | 0.573 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 9.2                    | 28.4            | 19.6 | 25.9 | 18.6 | 23.9 | 17.8 | 20.4                | 26.6 | 19.3 | 24.6 | 18.3 | 22.8 | 3.7                  | 110  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 18.1                            | 13.1 | 22.5 | 17.2 | 12.3 | 21.5 | 16.4 | 11.7 | 20.7 | 58.6          | 26.6 | 54.7 | 24.8 | 51.7 | 22.9 | 25.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (d)                        | (e)   | (f)                | (g) | (h)                                                | (i)   | (j)        | (k)   | (l)        | (m)   | (n)        | (o)   | (p)  | (q) |
| (4) 10.4          | 8.9  | 7.7 | DB                         | -11.7 | 32.3               | 2.9 | 1.9                                                | -13.9 | 33.6       | -15.6 | 34.8       | -17.2 | 35.8       | -19.4 | 37.2 | (4) |
| (5)               |      |     | WB                         | -11.8 | 22.8               | 2.9 | 1.5                                                | -13.9 | 23.9       | -15.6 | 24.8       | -17.2 | 25.6       | -19.3 | 26.7 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 9.2   | 2.0   | 2.7   | 4.6   | 8.2   | 11.9  | 14.5  | 17.1  | 17.6  | 14.0  | 9.7   | 5.4   | 2.0  | (6)  |
| (7)  |                                                                                  | DBStd       | 6.62  | 4.35  | 4.19  | 3.27  | 3.70  | 3.38  | 3.21  | 3.15  | 3.25  | 2.73  | 3.46  | 3.32  | 4.29 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 1156  | 247   | 205   | 169   | 78    | 18    | 1     | 0     | 0     | 2     | 47    | 140   | 250  | (8)  |
| (9)  |                                                                                  | HDD18.3     | 3411  | 505   | 437   | 426   | 304   | 202   | 124   | 63    | 54    | 134   | 266   | 388   | 507  | (9)  |
| (10) |                                                                                  | CDD10.0     | 856   | 0     | 1     | 2     | 24    | 77    | 135   | 220   | 235   | 121   | 39    | 2     | 1    | (10) |
| (11) |                                                                                  | CDD18.3     | 70    | 0     | 0     | 0     | 0     | 2     | 9     | 26    | 31    | 3     | 0     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 568   | 0     | 0     | 0     | 3     | 24    | 87    | 207   | 221   | 26    | 0     | 0     | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 151   | 0     | 0     | 0     | 0     | 4     | 26    | 57    | 62    | 2     | 0     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 3.5   | 4.1   | 4.4   | 4.2   | 3.4   | 3.0   | 3.1   | 3.0   | 2.9   | 3.0   | 3.5   | 3.5   | 3.6  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 835   | 66    | 53    | 49    | 41    | 56    | 74    | 91    | 87    | 85    | 77    | 78    | 78   | (15) |
| (16) |                                                                                  | PrecMax     | 1156  | 141   | 127   | 99    | 111   | 130   | 159   | 164   | 193   | 243   | 200   | 163   | 206  | (16) |
| (17) |                                                                                  | PrecMin     | 540   | 4     | 25    | 10    | 6     | 12    | 11    | 39    | 28    | 19    | 19    | 8     | 25   | (17) |
| (18) |                                                                                  | PrecStd     | 130   | 35    | 26    | 28    | 23    | 25    | 36    | 39    | 36    | 50    | 38    | 40    | 43   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 11.6  | 13.2  | 16.5  | 22.4  | 25.9  | 30.8  | 31.2  | 31.4  | 26.2  | 19.6  | 13.7  | 12.5 | (19) |
| (20) |                                                                                  |             | MCWB  | 9.6   | 10.1  | 10.7  | 14.5  | 16.9  | 20.8  | 21.2  | 20.3  | 18.4  | 15.6  | 12.2  | 10.9 | (20) |
| (21) |                                                                                  | 2%          | DB    | 10.1  | 11.2  | 13.3  | 19.2  | 22.9  | 25.8  | 28.1  | 28.3  | 23.1  | 17.1  | 12.4  | 11.0 | (21) |
| (22) |                                                                                  |             | MCWB  | 8.6   | 9.0   | 9.6   | 11.9  | 15.9  | 18.2  | 19.7  | 19.5  | 17.1  | 14.4  | 10.8  | 10.0 | (22) |
| (23) |                                                                                  | 5%          | DB    | 8.8   | 9.9   | 11.4  | 16.9  | 20.4  | 22.7  | 25.4  | 25.7  | 20.4  | 15.8  | 11.2  | 9.2  | (23) |
| (24) |                                                                                  |             | MCWB  | 7.4   | 7.9   | 8.7   | 11.5  | 14.4  | 16.9  | 18.7  | 19.0  | 15.7  | 13.5  | 9.4   | 8.1  | (24) |
| (25) |                                                                                  | 10%         | DB    | 7.3   | 8.3   | 10.0  | 14.5  | 18.2  | 20.3  | 23.2  | 23.4  | 18.4  | 14.7  | 10.0  | 8.0  | (25) |
| (26) |                                                                                  |             | MCWB  | 6.1   | 6.9   | 8.0   | 10.3  | 13.5  | 15.6  | 17.6  | 17.9  | 14.6  | 12.7  | 8.7   | 7.1  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 10.0  | 10.7  | 11.7  | 14.7  | 18.7  | 21.1  | 21.8  | 22.0  | 19.1  | 16.3  | 12.7  | 11.2 | (27) |
| (28) |                                                                                  |             | MCDWB | 11.2  | 12.5  | 14.7  | 20.1  | 24.4  | 29.0  | 28.7  | 28.4  | 25.2  | 18.8  | 13.5  | 12.0 | (28) |
| (29) |                                                                                  | 2%          | WB    | 8.7   | 9.4   | 10.2  | 13.1  | 16.6  | 19.0  | 20.5  | 20.3  | 17.7  | 14.8  | 11.1  | 9.9  | (29) |
| (30) |                                                                                  |             | MCDWB | 9.9   | 10.8  | 12.3  | 18.0  | 21.7  | 24.7  | 26.7  | 25.9  | 22.4  | 16.2  | 12.2  | 11.0 | (30) |
| (31) |                                                                                  | 5%          | WB    | 7.3   | 8.1   | 9.2   | 11.9  | 15.1  | 17.5  | 19.2  | 19.2  | 16.2  | 14.0  | 9.8   | 8.2  | (31) |
| (32) |                                                                                  |             | MCDWB | 8.5   | 9.6   | 11.0  | 15.9  | 19.2  | 21.5  | 24.5  | 24.2  | 19.3  | 15.4  | 10.9  | 9.0  | (32) |
| (33) |                                                                                  | 10%         | WB    | 6.2   | 7.0   | 8.0   | 10.7  | 13.8  | 16.3  | 18.0  | 18.4  | 15.2  | 13.0  | 8.7   | 7.0  | (33) |
| (34) |                                                                                  |             | MCDWB | 7.2   | 8.2   | 9.9   | 14.1  | 17.4  | 19.7  | 22.3  | 22.7  | 17.6  | 14.4  | 9.8   | 8.0  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 4.6   | 5.8   | 6.9   | 8.9   | 9.2   | 9.2   | 8.8   | 9.2   | 8.2   | 7.0   | 5.4   | 4.7  | (35) |
| (36) |                                                                                  |             | MCDBR | 5.4   | 7.3   | 9.7   | 12.6  | 13.2  | 13.2  | 12.7  | 12.4  | 11.1  | 7.4   | 6.0   | 5.2  | (36) |
| (37) |                                                                                  |             | MCWBR | 4.6   | 5.4   | 6.4   | 7.0   | 7.1   | 7.1   | 5.9   | 5.7   | 5.9   | 5.1   | 4.8   | 4.9  | (37) |
| (38) |                                                                                  | 5% WB       | MCDWR | 5.0   | 5.4   | 8.6   | 11.2  | 11.9  | 12.2  | 11.9  | 11.0  | 9.9   | 7.0   | 5.7   | 5.3  | (38) |
| (39) |                                                                                  |             | MCWBR | 4.6   | 4.4   | 6.1   | 6.7   | 6.8   | 7.1   | 5.9   | 5.6   | 5.6   | 5.2   | 5.0   | 5.0  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.327 | 0.346 | 0.375 | 0.408 | 0.403 | 0.402 | 0.417 | 0.411 | 0.387 | 0.362 | 0.341 | 0.320 | (40) |      |
| (41) |                                                                                  | taud        | 2.313 | 2.327 | 2.266 | 2.231 | 2.285 | 2.314 | 2.288 | 2.315 | 2.369 | 2.398 | 2.345 | 2.305 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 629   | 744   | 798   | 819   | 845   | 849   | 827   | 809   | 780   | 717   | 606   | 559   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 60    | 81    | 105   | 124   | 123   | 121   | 122   | 113   | 95    | 75    | 59    | 51    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.57  | 1.23  | 2.45  | 4.08  | 5.09  | 5.33  | 5.06  | 4.20  | 2.99  | 1.66  | 0.71  | 0.43  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.06  | 0.18  | 0.25  | 0.43  | 0.47  | 0.40  | 0.61  | 0.33  | 0.32  | 0.17  | 0.07  | 0.04  | (45) |      |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |  |  |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|--|--|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |  |  |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |  |  |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |  |  |
| (47) Regional (49 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -131    | N/A     | N/A     |  |  |

Nomenclature: See separate page